Nokia

NOK | NYSE ADR; underlying Nasdaq Helsinki NOKIA | 2026-10-07

Q4 2027 model ceiling: USD 13.37

Analytical Base 39/60 = 65 | Diversified network supplier with a temporary optical-capacity bottleneck

20x rolling Q+next3 IFRS operating profit after1% common-attribution reserve, multiplied by1.14 USD/EUR and divided by window-end ADS-equivalent diluted shares; no excess-cash bonus.

Coverage initiated at the owner's request. No holding was established by this research. Existing portfolio Base overrides remain protected.

All operating values are in EUR m and shares in millions unless stated otherwise. Forecasts are analytical assumptions; issuer guidance and actual results are identified separately.

Writing-date snapshot: USD 10.60 per share (2026-10-02 issuer IR quote; dated reference); 5,598.645 million common shares (2026-06-30 issuer results); estimated equity capitalization USD 59,345.6 million. Quote and disclosed share count have different dates; capitalization is their stated-date product, not a live market feed. This dated common-share snapshot is separate from forecast diluted shares.

Decision in the model

Operating profit is in EUR million; equity value is translated at 1.14 USD per EUR, with one ordinary share per ADS.

Active horizon	Earnings window	F4Q OP	Multiple	Diluted shares	Ceiling
Q4 2027	Q4 2027 through Q3 2028	3,362.0	20x	5,734.000	13.37

The ceiling is a model valuation, not a guaranteed trading price or a literal net-income P/E. Operating branches are weighted once. Where debt, preferred claims, conversion or equity floors make value nonlinear, the model values each coherent operating and funding state first, then weights those common-share outcomes.

Economic scope and matching claims

IFRS operating profit includes production D&A, R&D/SG&A/SBC, acquired-intangible amortization, restructuring and integration. Nonoperating venture gains excluded.. 20x rolling Q+next3 IFRS operating profit after1% common-attribution reserve, multiplied by1.14 USD/EUR and divided by window-end ADS-equivalent diluted shares; no excess-cash bonus.

Financial claims and diluted shares follow the company-specific schedule below. Debt, preferred and minority claims are counted once. All financial model tables use EUR m unless a different unit is stated.

Thesis

Research date: October 7, 2026. Nokia Oyj, NYSE NOK ADS; one ADS represents one ordinary share. Operating amounts are EUR million; ceiling values are USD per ADS.

The Q4 2027 model ceiling is **$13.37 per ADS**, using **20x forward-four-quarter IFRS operating profit**, a 1% common-profit attribution reserve and **$1.14 per euro**. The window runs from Q4 2027 through Q3 2028. Analytical Base is **65/100, or 39/60**. Nokia is a diversified network beneficiary with a temporary optical-capacity bottleneck; its entire portfolio does not qualify as an irreplaceable optical choke point.

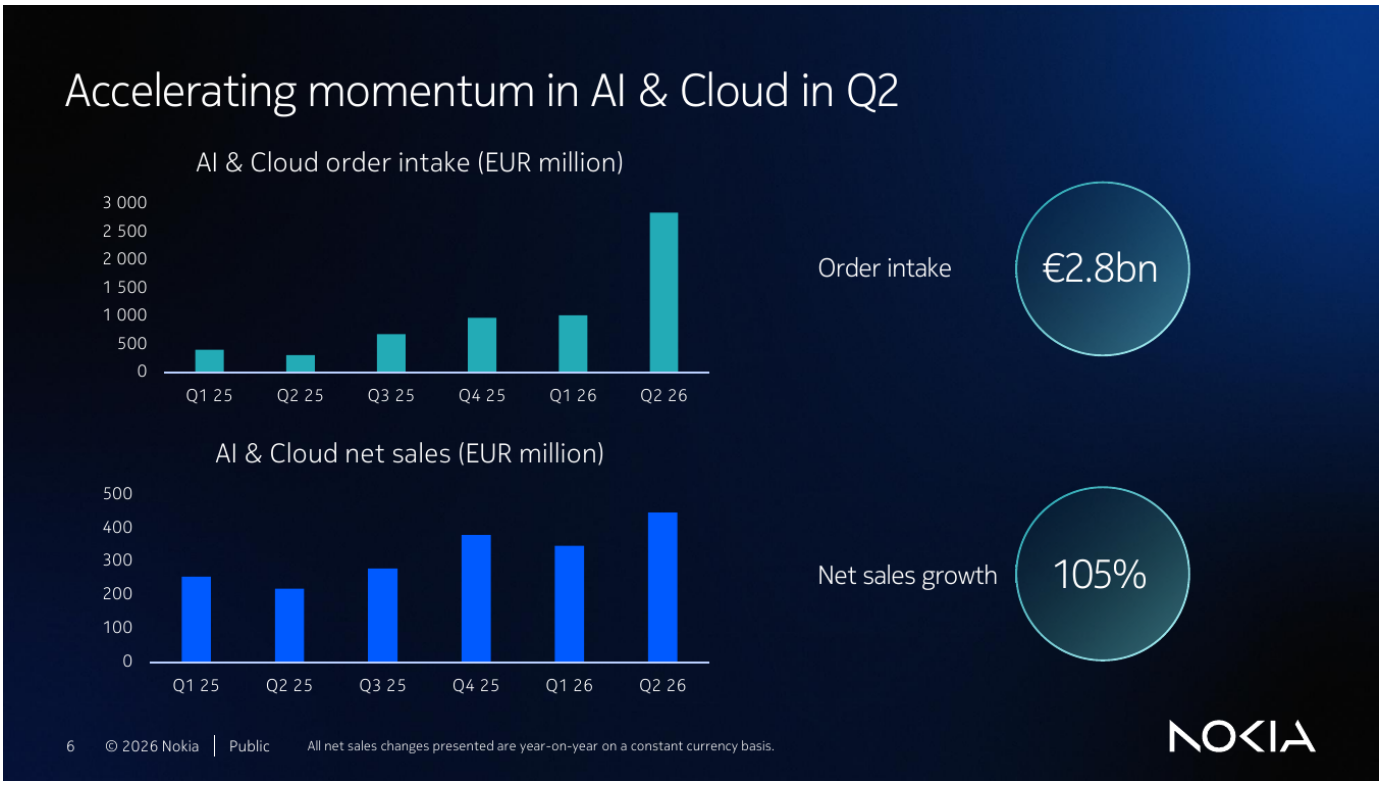
The investable change is measurable order conversion. Nokia has optical systems, coherent DSPs, InP photonic integration, packaging and customer relationships under one roof. That gives it several ways to improve delivered cost, power and capacity per link. The remaining test is whether qualified output and external invoices grow faster than manufacturing, engineering, stock compensation and integration expense.

Orders are not immediate sales

Q2 AI and cloud orders were €2.8bn against €446m recognized sales. Management expected around half of those orders to convert over the next twelve months. The remainder extends beyond that period; it is not another €2.8bn quarterly run rate. April management clarified that its reported orders are purchase orders with delivery dates, rather than the entire value of multiyear framework agreements. July management nonetheless cautioned about lumpiness and longer commitments. Q2 report, April management discussion.

The order book contains both optical and IP equipment, weighted toward optical according to the July CEO response. It is also a customer-category view of the same revenue that appears in product segments. Adding it to Optical Networks and IP Networks would count the same sales twice. The reference forecast already includes substantial conversion of existing orders, increasing unit volumes and the normal San Jose ramp.

The useful monitoring series is accepted sales, order delivery dates, gross profit and working capital together. A growing backlog accompanied by an inability to qualify products can consume cash without generating the modeled profit. More orders from the same underlying program are not independent evidence that its delivery probability increased.



Original issuer exhibit | AI/cloud orders and sales are different series; around half of Q2 orders convert in twelve months.

The optical wallet and the engineering advantage

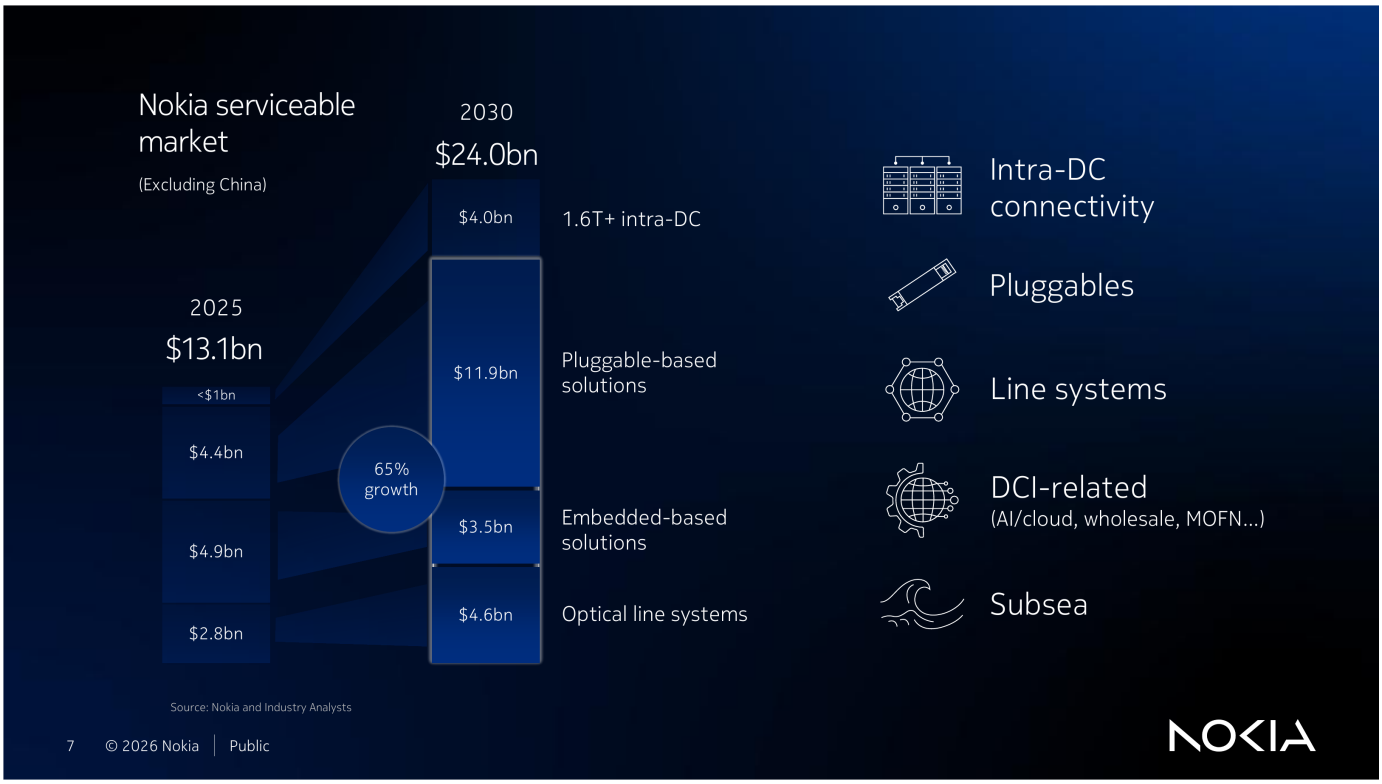
The March OFC deck identifies a $24bn 2030 serviceable optical market excluding China, comprising $4bn intra-data-center connectivity, $11.9bn pluggable-based solutions, $3.5bn embedded solutions and $4.6bn optical line systems. Its $13.1bn 2025 total includes less than $1bn intra-DC. The pictured 65% growth is consistent with the conventional $12.1bn subtotal growing to $20bn; it is not the growth rate from the displayed $13.1bn total to $24bn. These categories are a company-defined serviceable market, not the framework's global transceiver market. Original OFC deck.

InP photonic integration can combine optical functions and reduce separate assembly and coupling operations. Coherent DSPs and suitable optics encode and recover signals over distances where an intensity-modulated short-reach product is not a substitute. As spectral efficiency approaches practical limits, distributing traffic over more fibers and reducing cost, power and space per useful bit can matter more than maximizing a single fiber's peak capacity. The relevant comparison is the complete accepted link at a given reach and error rate.

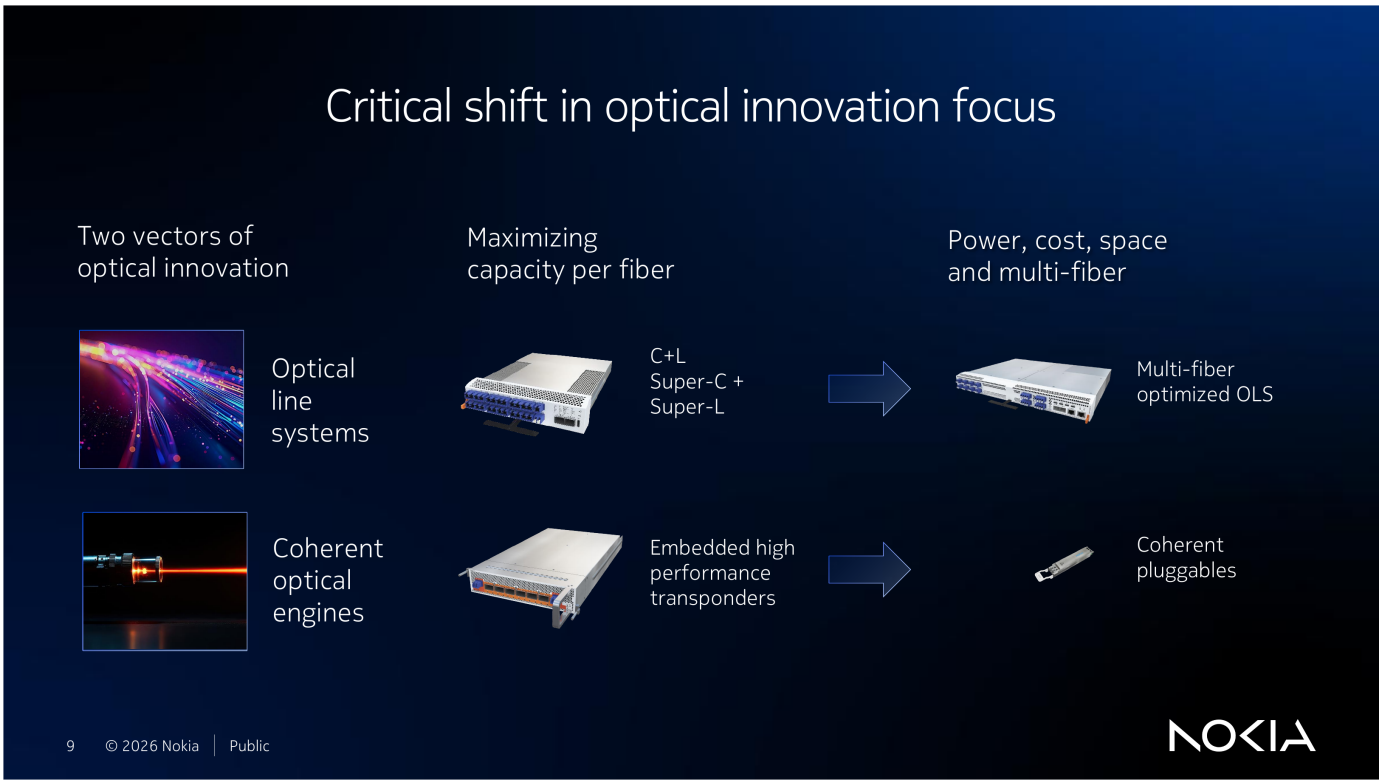
ICE-D addresses intra-DC architectures including fully retimed, half-retimed and linear-drive optics. LPO removes module DSP functions but shifts demands onto host signal integrity and link qualification. It is not identical to coherent-lite, and neither is interchangeable with a high-power external laser serving CPO. This model books Nokia's external system/component revenue once; semiconductor output consumed inside its own systems is not extra sales. ICE-D technology.

The reference optical sales path is €4.6bn, €5.8bn, €6.9bn and €7.7bn in 2027-30. At $1.14 per euro, the 2030 reference is about 36.6% of the issuer's dated $24bn serviceable market; the active output and price branches raise it to about 40.5%. That is a demanding capture scenario, above a rough 2025 pro-forma share near 28%. It requires competitive gains and qualified supply, not simply faster industry growth. No portion of the full telecom or AI infrastructure capex pool is assigned automatically to Nokia.

Adjacent demand comes from cloud interconnect, routed coherent links, autonomous networks and AI-RAN. Physical AI could create more machine traffic and distributed edge inference, but additional traffic has to become a paid optical, routing or software order. The model does not add a separate robots, quantum or space revenue pool without a product and customer bridge.



Original issuer exhibit | Issuer optical serviceable market, excluding China; categories overlap neither IP nor mobile revenue.



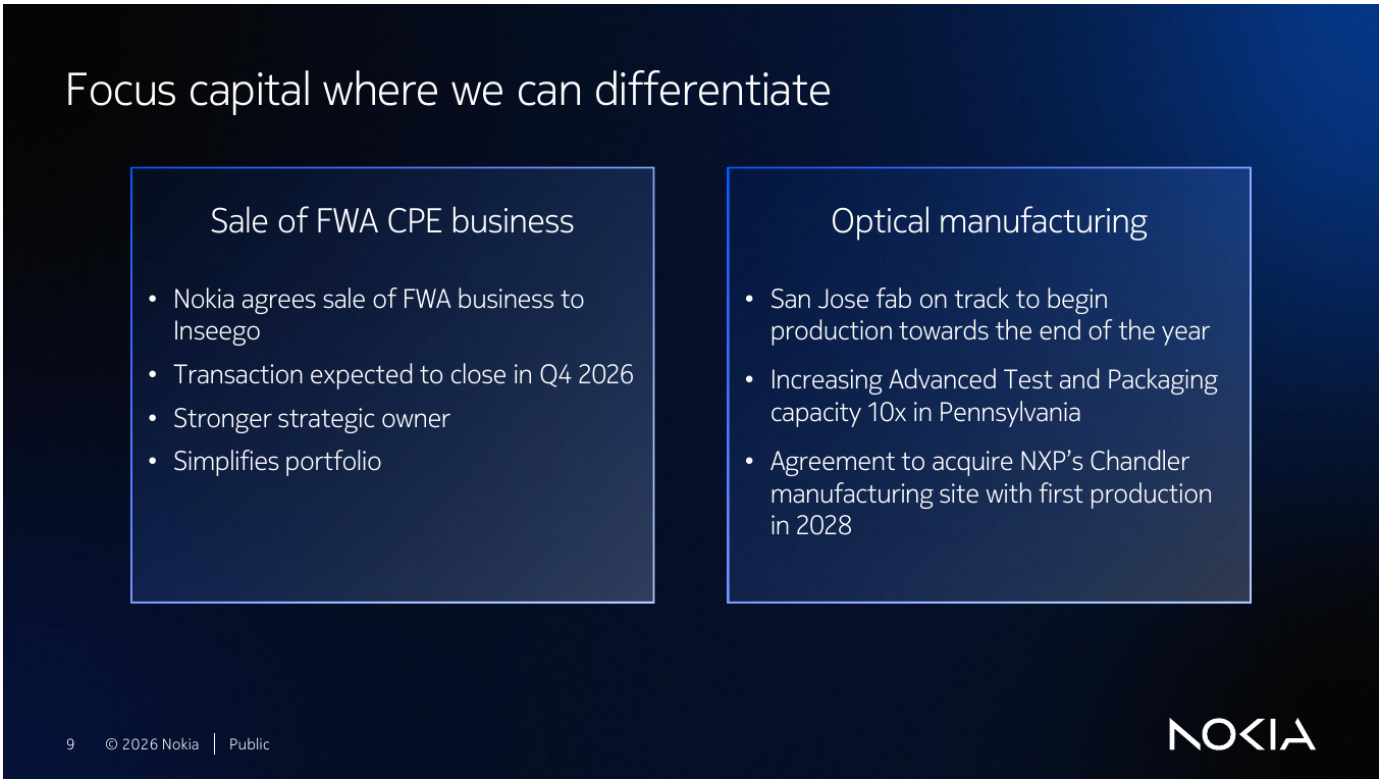
Original issuer exhibit | System design shifts from maximizing one fiber to power, space and multi-fiber economics.

Qualified capacity is the bottleneck

San Jose was processing test wafers in July, with production targeted near year-end 2026 and a 2027 volume ramp. Pennsylvania's test and packaging expansion begins in Q3 2026. The disclosed 10x increase applies to that facility's advanced test and packaging capacity; it does not multiply company sales, optical wafer output or accepted systems by ten. Yield, die size, test time, DSP availability, assembly and customer acceptance remain separate constraints.

Chandler has three different milestones. The financial report describes initial leasing in early 2027 and acquisition closing in Q1 2029. The presentation says first production in 2028, while the CEO's supply discussion describes the incremental fab ramp as likely 2029 at the earliest. The model treats 2028 as limited initial qualification and 2029 as the meaningful capacity contribution. It does not assume ownership at the lease date or full output at first production. Purchase price, required conversion expenditure and accepted output are still undisclosed. July management discussion.

Customers may pay to obtain a qualified link earlier because delayed compute or network deployment is expensive. But April management explicitly said the growth forecast was volume-led, with falling optical cost curves, rather than a general price-increase story. The active premium therefore has only 30% probability, affects 25% of reference optical sales and assumes a 2% premium in 2027-28. Higher memory/DSP/PCB costs are an offset, not evidence that every increase in invoiced equipment revenue becomes profit.



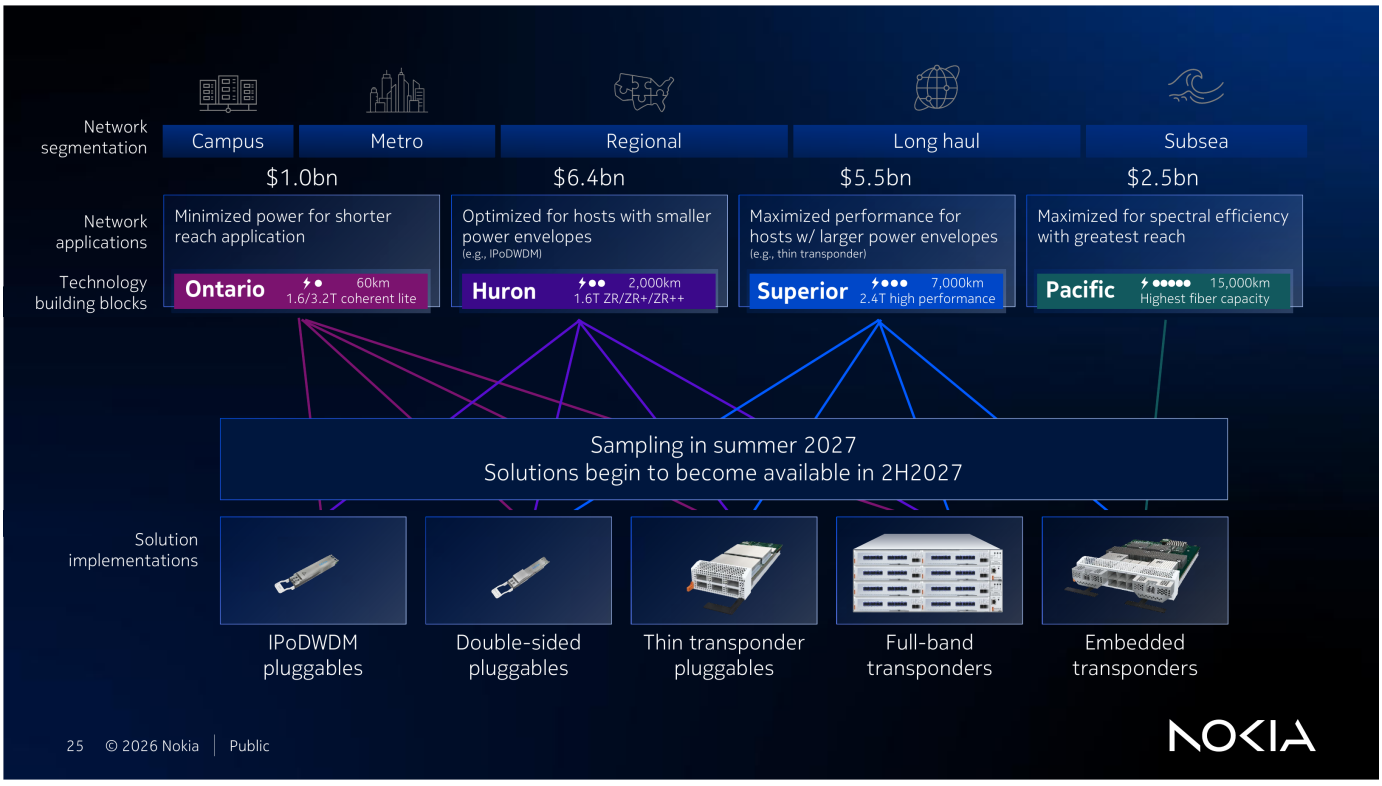
Original issuer exhibit | San Jose, Pennsylvania and Chandler are distinct stages and dates;10x packaging is not10x company revenue.

Roadmap, substitution and competition

Ontario targets shorter coherent-lite reach, Huron pluggable ZR-type applications, Superior high-performance transponders and Pacific maximum-reach/spectral-efficiency applications. The four-DSP roadmap preserves work already underway in the formerly separate Nokia and Infinera teams. It does not mean four incremental businesses without corresponding engineering cost. The OFC schedule shows sampling in summer 2027 and solutions beginning to become available in H2 2027; July management puts customer trials in 2027 and commercial availability toward year-end.

Optical systems face Ciena, Huawei and other system suppliers; components and pluggables face established optical specialists. Routing and switching face Cisco, Arista, Juniper/HPE and hyperscaler architectures. Radio faces Ericsson, Samsung and other country-dependent suppliers, including open and virtualized approaches. Nokia's annual risk disclosure explicitly describes equipment price erosion, large-customer bargaining, acceptance timing and incumbent competition in data-center entry. Vertical integration improves coordination; it does not remove those competitors. 2025 Form 20-F.

AI-RAN is a credible expansion opportunity but also a business-model transition. Ten publicly discussed pilot customers do not establish ten paid production contracts. Moving baseband processing to general-purpose silicon may lower some custom-silicon investment and enable software upgrades, while also transferring hardware economics toward GPU suppliers. The active model uses a 35% probability on net new Nokia sales after replacement of conventional RAN revenue, with heavy incremental engineering/support costs.



Original issuer exhibit | Sampling and initial availability do not establish fully qualified volume output.

Financial anchor and fully expensed profit

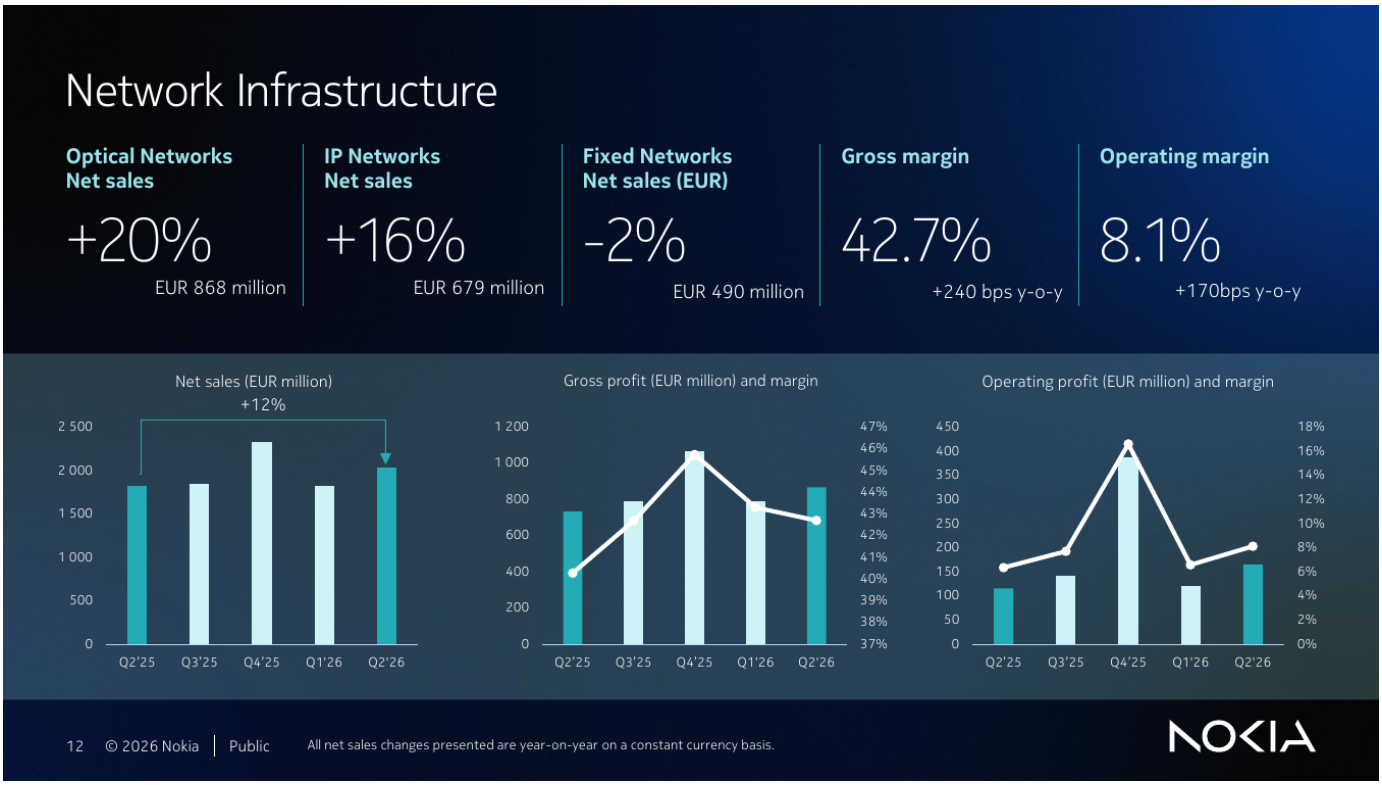
The last complete quarter is Q2 2026: sales €4,815m, reported operating loss €50m and comparable operating profit €434m. The €484m difference includes €390m restructuring, €46m acquired-asset D&A, €30m impairment/write-offs and €14m integration costs, with other small reconciliation items. H1 continuing sales were €9,248m, reported OP €33m and comparable OP €735m. Venture-fund revaluation benefits below operating profit are excluded from the valuation.

July guidance is €2.1-2.6bn comparable OP, with management tracking somewhat above the midpoint. The model calibrates to €2.4bn comparable OP, then deducts the costs needed to produce an IFRS estimate. Q3 sales of €5,040m are 4.7% above Q2, within the 3-7% sequential guide; modeled comparable OP is €435m. Q4 comparable OP is €1,230m, reflecting seasonal software and product deliveries. H2 restructuring charges of €254m complete the €800m full-year guide. Q3 statutory OP is €220m and Q4 €1,069m before the small common-attribution reserve. These quarters remain forecasts pending October 22 results.

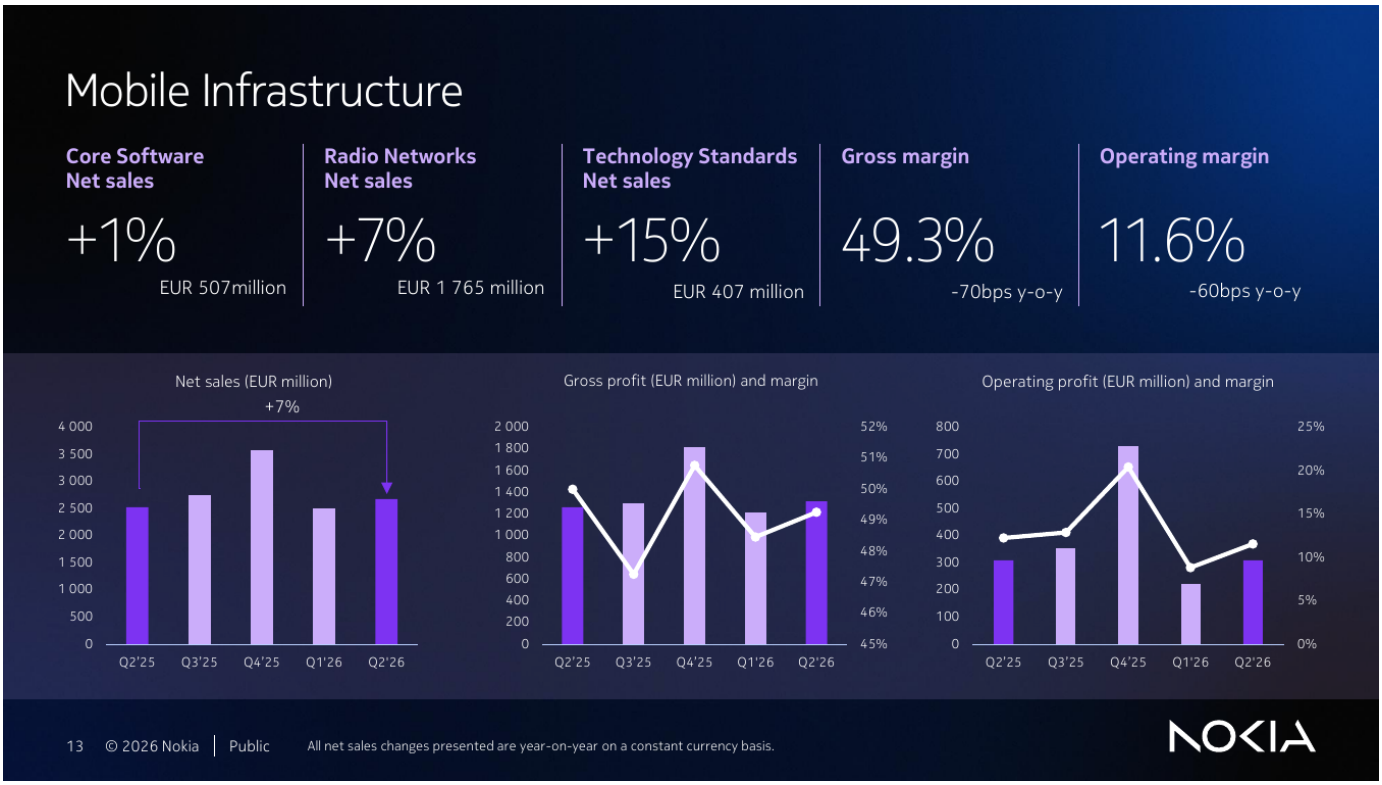
Seven revenue vectors retain separate margins: optical; IP; fixed networks; radio; core software; technology licensing; remaining portfolio. The first three make up Network Infrastructure. Radio, core and licensing make up Mobile Infrastructure. Product gross margins are modeling allocations calibrated to disclosed segment data, not reported product margins. In particular, Q3 mobile mix produces about 45.5% gross margin, consistent with management's 44-46% indication.

From 2027, reference R&D is 22.5% of revenue, falling to 20.2% by 2030, while SG&A/net other operating costs fall from 11.2% to 10.2%. These allocations include SBC. Production D&A is in gross margin; acquired-intangible amortization remains a separate €175m/165m/150m/135m annual charge. Restructuring remains €240m/160m/120m/120m, and integration/impairment reserves remain €45m/30m/20m/20m. Savings are therefore not modeled as free or instantaneous.

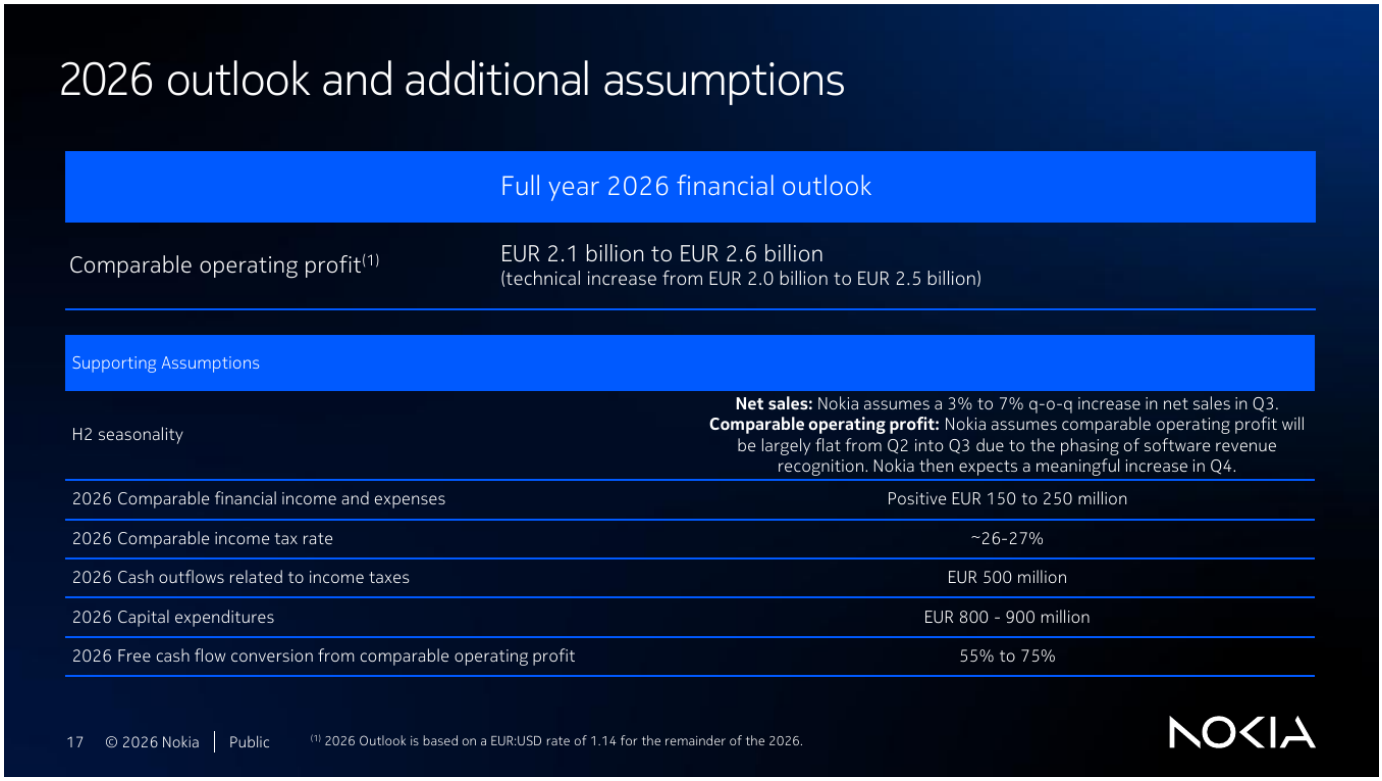
The group margin improvement is ambitious and conditional on scale, richer optical mix and operating leverage. The long-range path is a company model, not new management guidance. An inability to improve gross margin while sustaining R&D would materially reduce it.



Original issuer exhibit | Network Infrastructure has disclosed segment margins; product-level margins in our build are assumptions.



Original issuer exhibit | Mobile and licensing remain a large share of group earnings; they do not inherit optical growth.



Original issuer exhibit | July guidance is comparable OP; the model deducts statutory reconciliation costs.

Probability-weighted operating bridge

Each increment is measured against the already growing reference. Probabilities are subjective model judgments. For Q4 2027 through Q3 2028, the fully costed branches contribute:

Branch	Probability	Weighted revenue, EURm	Weighted common OP, EURm
Higher qualified optical output	55%	352.00	97.57
Repriceable optical premium	30%	8.18	4.45
Net AI-RAN conversion	35%	77.00	11.43
Residual unrecovered input costs	40%	0.00	-36.86

The output endpoint adds €400m/800m/1,200m/1,500m annual optical invoices in 2027-30 above the reference. A €20,000 external-system-equivalent proxy corresponds to 20k/40k/60k/75k accepted equivalents; this is not a disclosed physical unit mix. Conditional incremental GP is 40%, after manufacturing D&A, yield and expediting, with 12% incremental R&D/selling expense. The branch prices the residual qualified-output opportunity, rather than adding the order book again.

The price branch retains 60% of incremental invoice value in gross profit and spends another 5% on commercial support. It applies only to reference units, so it does not inflate the volume branch a second time. The premium fades to 1.5% in 2029 and 1% in 2030. The input-cost branch charges a separate residual 0.8% of reference Network Infrastructure sales; ordinary manufacturing and the output branch's expediting are already elsewhere.

The AI-RAN endpoint adds net €100m/300m/600m/900m annual sales, equivalent to 10/20/30/30 programs at €10m/15m/20m/30m annual net Nokia invoices. At 35% GP and 20% incremental engineering/support, it earns a

15% conditional OP margin. These are bounded commercial proxies, not announced contract values. No GPU supplier revenue or displaced legacy RAN invoice is counted twice.

Reference window common OP is €3,285.447m; active OP is €3,362.046m. The valuation is **€3,362.046m x 20 x $1.14/€ ÷ 5,734m ADS equivalents = $13.3684**. The no-increment control is $13.0639. A 10% change in the valuation FX input moves the dollar ceiling 10% before any operating-currency effects; it is a diagnostic, not a forecast that Nokia's profit is FX-neutral.

Funding, shares and the ADR bridge

June cash and equivalents were €4,348m and current interest-bearing investments €484m. The funding test begins with €4,832m, excluding noncurrent investments, pension assets and deferred-tax assets. Interest-bearing debt was €2,400m and leases €959m. The issuer's €2,776m net-cash headline uses a broader investment definition and excludes leases; it must not be mistaken for unencumbered excess equity value.

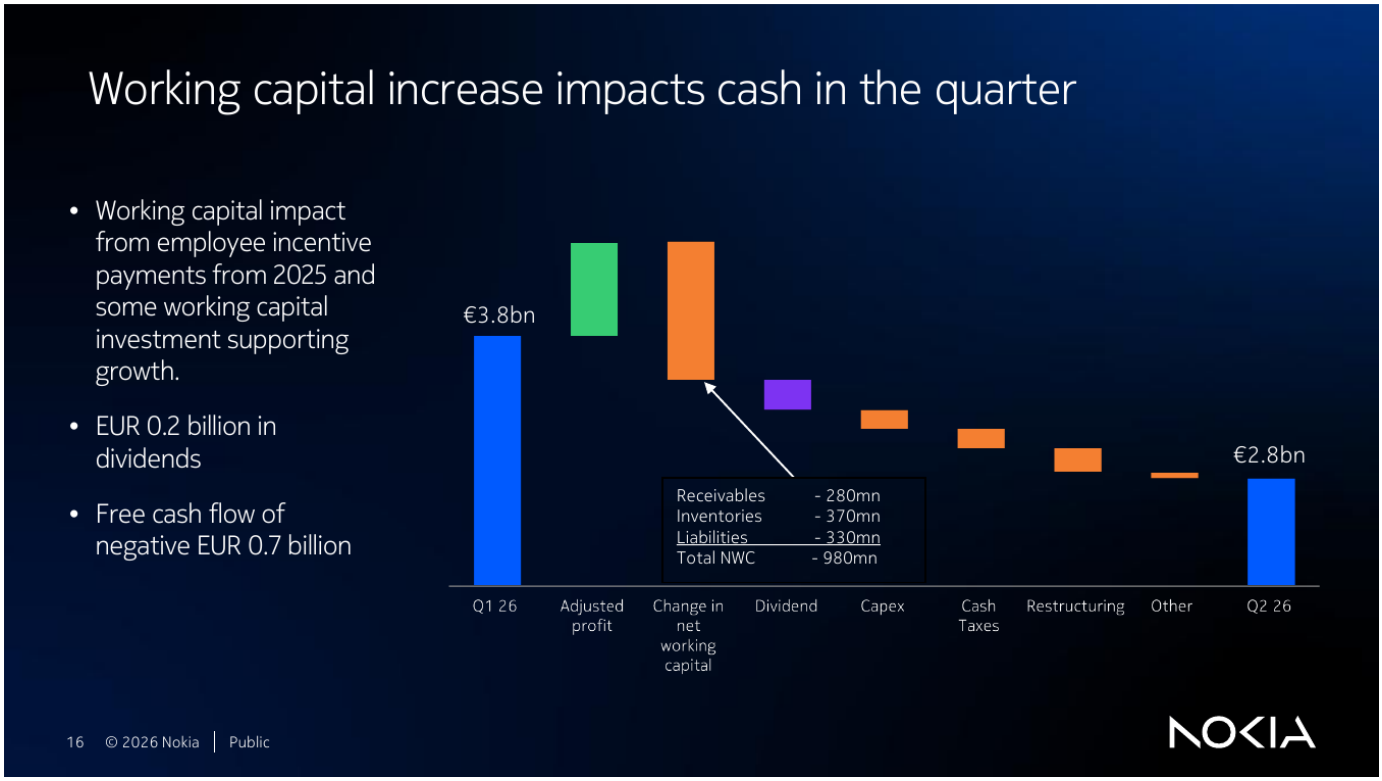
Known bond maturities include approximately €431m in June 2027, €65m in January 2028 and €181m in March 2029. Committed facilities comprise a €1,500m revolver and two €435m EIB facilities; they are availability, not cash already received. No facility draw is required in the reference or selected active cash path. Both maintain a minimum modeled cash balance around €4,613m over the forecast. The model gives no bonus for accumulated surplus cash or pension assets.

The cash bridge includes operating profit, cash taxes, D&A and SBC add-backs, capex, growth working capital, interest, leases, dividends and scheduled debt repayment. SBC remains fully expensed in OP; its cash add-back is paired with growing diluted shares. Capex rises from the €800-900m 2026 guidance context to model assumptions of €1.15bn, €1.30bn, €1.50bn and €1.30bn in 2027-30. A €500m Q1 2029 Chandler acquisition reserve is explicitly assumed because price is undisclosed; it is not a statement of agreed consideration.

A severe funding stress reduces OP by 35%, increases capex 50%, adds €1.5bn initial working capital, raises Chandler consideration to €1.5bn and retains dividends. Minimum cash falls to **negative €1.17bn** before new financing. That amount is below the currently disclosed aggregate facilities, but refinancing availability, covenants, spending changes and dividend restraint still need verification when the actual acquisition economics emerge. The stress is not described as self-funded. No future equity raise or undisclosed asset sale is assumed in the active case.

Issued shares at June were 5,742.240m; treasury shares 143.595m; outstanding common shares 5,598.645m. The model begins at 5,670m diluted ADS equivalents, including a 71.355m reserve above outstanding, and adds 8m per quarter for net stock settlement. It reaches 5,734m at the active window end. The 2025 20-F verifies one ADS represents one ordinary share. NVIDIA's earlier equity issuance is already inside the issued-share base and is not added again.

The 1% OP reserve for noncontrolling interests is a conservative profit attribution proxy against small reported NCI earnings; €97m book NCI is not deducted as if it were a debt obligation. Any future minority restructuring requires a subsidiary-level revision. $1.14 per euro is an explicit valuation convention drawn from July guidance, rather than an assertion about current spot FX.



Original issuer exhibit | The June working-capital outflow is real cash usage; neither pension assets nor venture marks fund capex.

Score, risks and cross-company tests

Category	Points	Reason
Core thesis	13/20	InP/PIC, DSP and system capability with real qualification barriers, inside a competitive diversified supplier
Cycle	7/10	Optical and switching ramp with mature telecom exposure
TAM expansion	8/10	DCI, intra-DC, network software and AI-RAN with different commercial pathways
Catalysts	4/5	October results, late-2026 fab/pilots and 2027 roadmap qualification
Macro	8/10	AI network traffic and trusted manufacturing; telco capital returns remain a constraint
Alpha	3/5	Product/order/IFRS reconciliation is useful; broad coverage limits information advantage
Risks	-4	Technical and supply execution; customer competition; restructuring and capital commitments; FX and geopolitics
Total	39/60	**65/100**

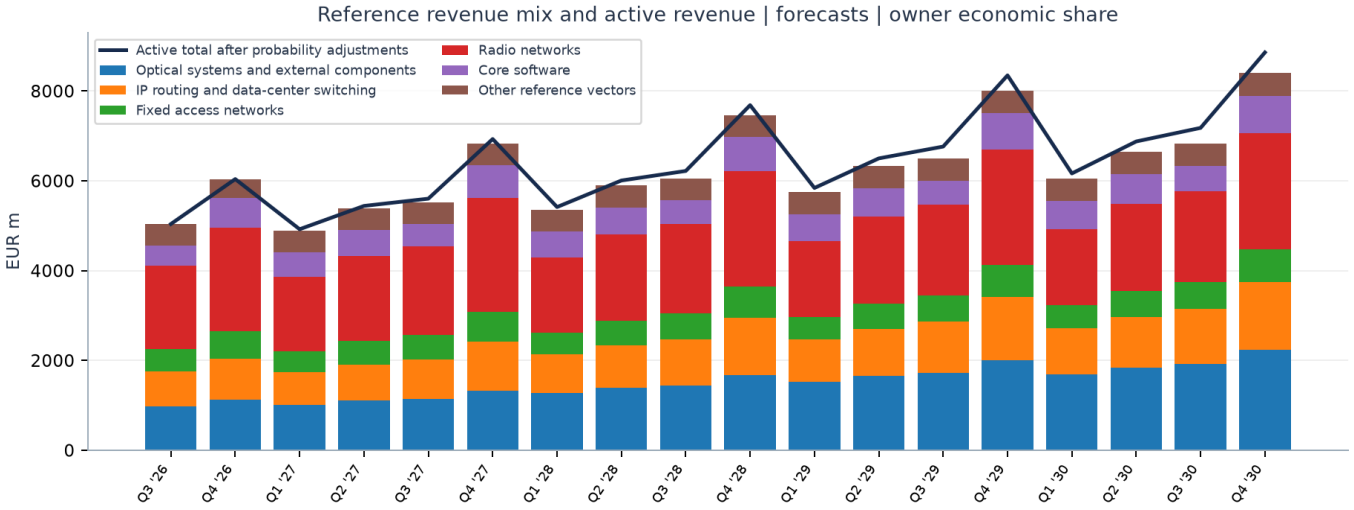
LITE and COHR can be suppliers, competitors or alternative optical architectures depending on the exact product. Nokia's internal InP expansion does not establish an external award to either. MRVL coherent silicon and Ciena systems are competitive comparisons, not incremental Nokia revenue evidence. NVDA participates in AI-RAN hardware; Nokia captures software/platform economics only where paid. These cross-watches should compare accepted yield, measured link power/reach, operator return on capital and customer sourcing.

Raise the qualified-output probability only when accepted external sales, yields and backlog conversion improve together. Lower it toward 30% after a two-quarter qualification delay. Raise the AI-RAN branch toward 50% on paid production deployments; lower it toward 15% if trials slip or customer economics fail. Remove the limited premium if contracts remain fixed and cost inflation absorbs realized increases. Revisit the capital reserve once Chandler terms and its acceptance schedule are disclosed. These tests are concrete enough to change the model before waiting for full-year confirmation.

Score arithmetic

Category	Points	Evidence / treatment
Core thesis	13	Vertical InP/PIC, DSP and system capability; competition prevents a whole-company choke point.
Cycle	7	Optical and AI switching ramp; mature telecom remains most sales.
TAM	8	DCI, intra-DC, software and AI-RAN expand addressable functions with explicit capture.
Catalysts	4	Q3 results October22; San Jose late2026 qualification; 2027 roadmaps and trials.
Macro	8	AI traffic supports demand, while telco returns and supplier inflation limit capture.
Alpha	3	Reconcile hardware systems, internal component transfers, order timing and reported IFRS costs.
Risk classes	-4	Four classes: technical/supply execution, customer/competitive concentration, restructuring/capital commitment, and FX/geopolitics.
Total	39/60	Rounded Base 65. Both artifact and new workbook row use this score; this is not a manual override.

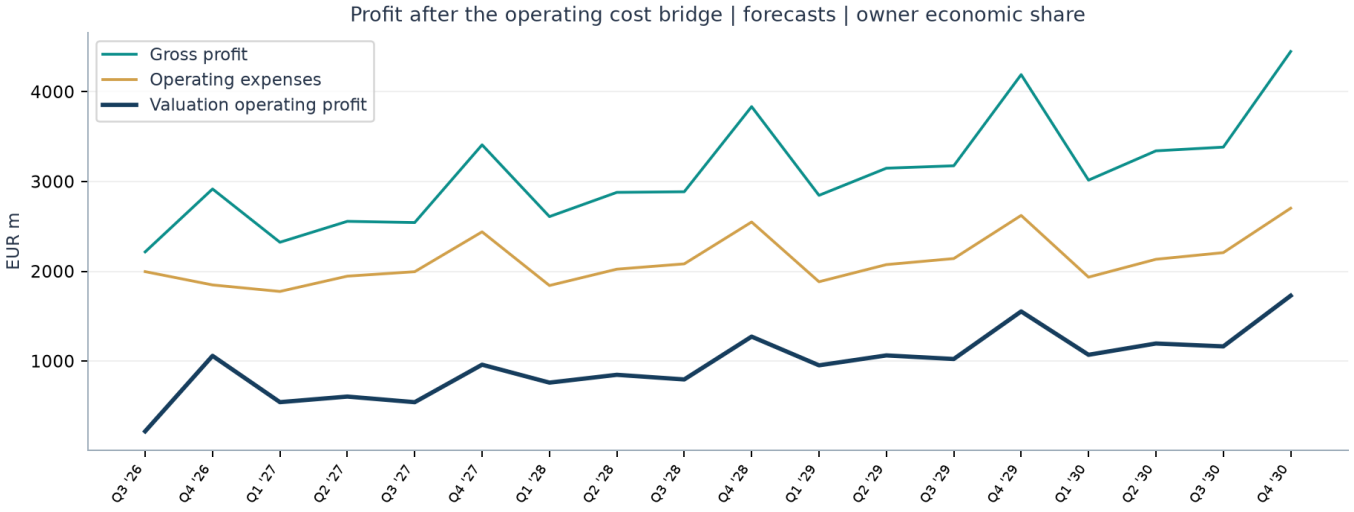
Revenue and business mix



Reference business vectors sum to the reference; the active total adds only probability-weighted net changes. Product allocations may be analytical proxies where issuers do not report product margins.

Quarter	Reference revenue	Weighted net change	Active revenue
Q3 2026	5,040.0	0.0	5,040.0
Q4 2026	6,040.0	0.0	6,040.0
Q1 2027	4,896.5	27.0	4,923.5
Q2 2027	5,389.5	52.7	5,442.2
Q3 2027	5,525.8	78.2	5,604.0
Q4 2027	6,828.2	104.0	6,932.3
Q1 2028	5,361.2	56.4	5,417.6
Q2 2028	5,897.4	111.1	6,008.5
Q3 2028	6,055.4	165.7	6,221.1
Q4 2028	7,465.9	220.5	7,686.5
Q1 2029	5,753.6	88.7	5,842.4
Q2 2029	6,325.4	175.9	6,501.3
Q3 2029	6,501.2	262.9	6,764.1
Q4 2029	7,999.7	350.3	8,350.0
Q1 2030	6,050.7	115.3	6,166.0
Q2 2030	6,648.9	229.4	6,878.3
Q3 2030	6,836.9	343.4	7,180.4
Q4 2030	8,403.5	457.7	8,861.1

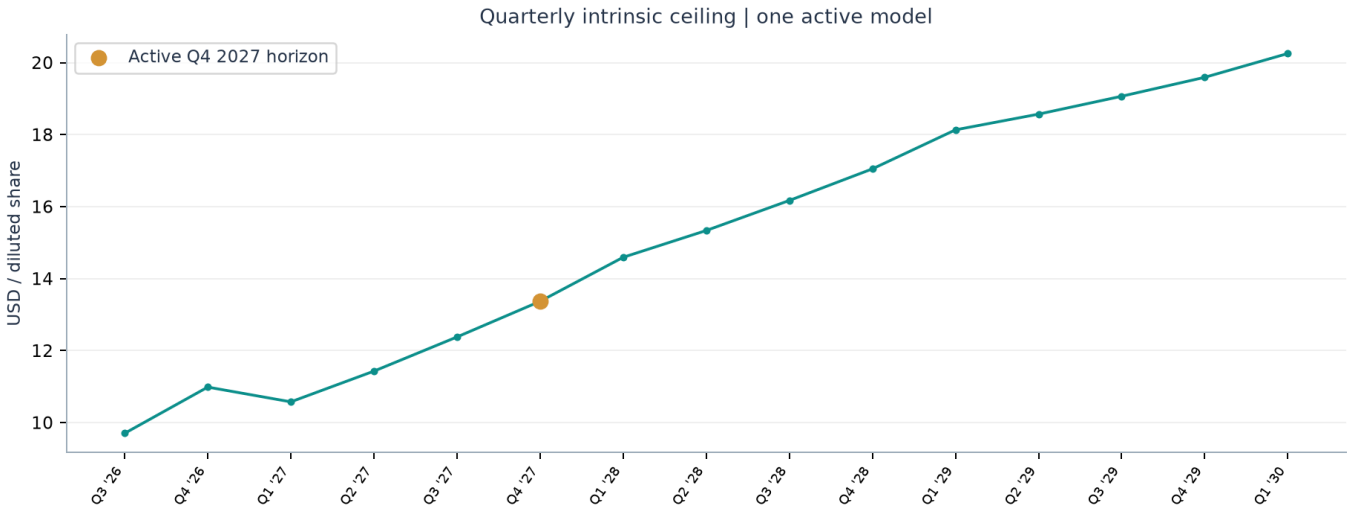
Gross profit, operating expenses and earnings



Operating profit equals gross profit less operating expenses and any separately identified attribution reserve. Figures are forecasts, not reported historical quarterly outcomes.

Quarter	Revenue	GP	Opex	Valuation OP	OP margin
Q3 2026	5,040.0	2,216.4	1,996.4	217.8	4.3%
Q4 2026	6,040.0	2,917.1	1,848.1	1,058.3	17.5%
Q1 2027	4,923.5	2,323.5	1,775.6	542.4	11.0%
Q2 2027	5,442.2	2,556.5	1,945.8	604.6	11.1%
Q3 2027	5,604.0	2,543.1	1,995.5	542.1	9.7%
Q4 2027	6,932.3	3,408.8	2,439.4	959.6	13.8%
Q1 2028	5,417.6	2,609.4	1,841.6	760.0	14.0%
Q2 2028	6,008.5	2,879.2	2,023.6	847.0	14.1%
Q3 2028	6,221.1	2,886.1	2,082.7	795.4	12.8%
Q4 2028	7,686.5	3,834.3	2,549.0	1,272.4	16.6%
Q1 2029	5,842.4	2,846.0	1,883.6	952.8	16.3%
Q2 2029	6,501.3	3,148.4	2,074.5	1,063.2	16.4%
Q3 2029	6,764.1	3,175.3	2,141.9	1,023.1	15.1%
Q4 2029	8,350.0	4,190.8	2,622.2	1,552.9	18.6%
Q1 2030	6,166.0	3,015.4	1,934.8	1,069.9	17.4%
Q2 2030	6,878.3	3,341.7	2,133.8	1,195.8	17.4%
Q3 2030	7,180.4	3,383.2	2,207.9	1,163.5	16.2%
Q4 2030	8,861.1	4,448.2	2,702.6	1,728.2	19.5%

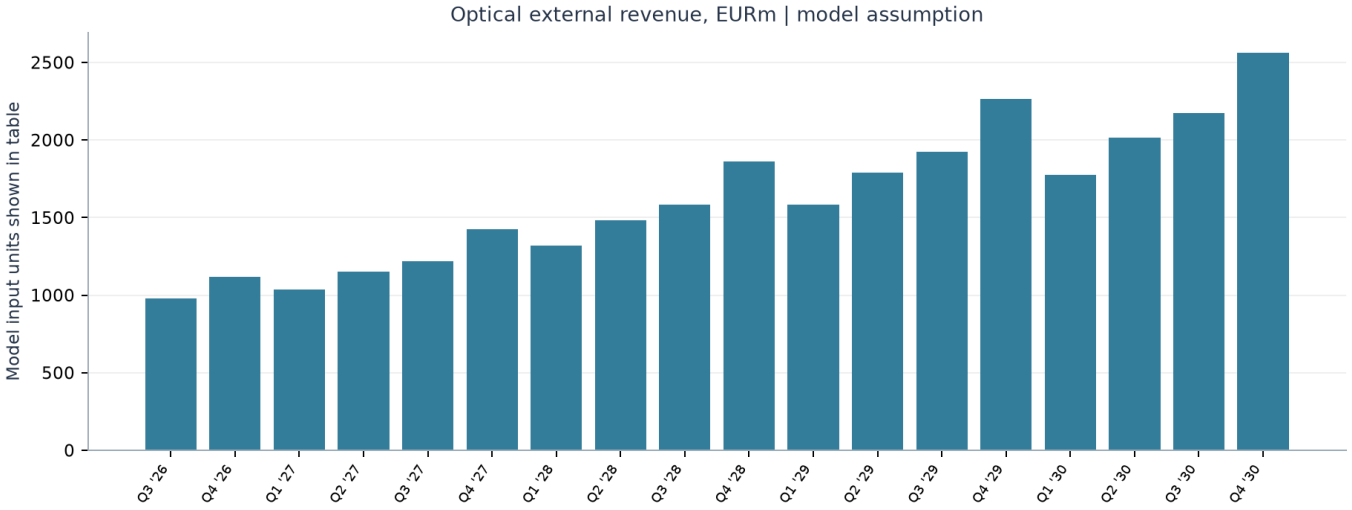
The complete ceiling path



Quarter Q prices Q plus the next three calendar quarters. All fifteen ceilings link to the same operating rows used in the workbook. Outer quarters have lower evidence visibility.

Quarter	Four-quarter earnings window	F4Q OP	Multiple	Shares	Ceiling
Q3 2026	Q3 2026 to Q2 2027	2,423.2	20x	5,694.000	9.70
Q4 2026	Q4 2026 to Q3 2027	2,747.4	20x	5,702.000	10.99
Q1 2027	Q1 2027 to Q4 2027	2,648.8	20x	5,710.000	10.58
Q2 2027	Q2 2027 to Q1 2028	2,866.4	20x	5,718.000	11.43
Q3 2027	Q3 2027 to Q2 2028	3,108.8	20x	5,726.000	12.38
Q4 2027	Q4 2027 to Q3 2028	3,362.0	20x	5,734.000	13.37
Q1 2028	Q1 2028 to Q4 2028	3,674.8	20x	5,742.000	14.59
Q2 2028	Q2 2028 to Q1 2029	3,867.6	20x	5,750.000	15.34
Q3 2028	Q3 2028 to Q2 2029	4,083.7	20x	5,758.000	16.17
Q4 2028	Q4 2028 to Q3 2029	4,311.4	20x	5,766.000	17.05
Q1 2029	Q1 2029 to Q4 2029	4,591.9	20x	5,774.000	18.13
Q2 2029	Q2 2029 to Q1 2030	4,709.0	20x	5,782.000	18.57
Q3 2029	Q3 2029 to Q2 2030	4,841.6	20x	5,790.000	19.07
Q4 2029	Q4 2029 to Q3 2030	4,982.0	20x	5,798.000	19.59
Q1 2030	Q1 2030 to Q4 2030	5,157.3	20x	5,806.000	20.25

The operating constraint behind the forecast



The driver series is a stated model assumption, not a disclosed commitment. It must be evaluated with the qualification, customer capture, cost and overlap discussion.

Quarter	Optical external revenue, EURm	Status
Q3 2026	980.000	Model
Q4 2026	1,120.000	Model
Q1 2027	1,035.518	Model
Q2 2027	1,149.656	Model
Q3 2027	1,217.725	Model
Q4 2027	1,424.001	Model
Q1 2028	1,321.914	Model
Q2 2028	1,482.088	Model
Q3 2028	1,584.175	Model
Q4 2028	1,860.523	Model
Q1 2029	1,585.708	Model
Q2 2029	1,789.863	Model
Q3 2029	1,924.941	Model
Q4 2029	2,267.251	Model
Q1 2030	1,777.771	Model
Q2 2030	2,014.386	Model
Q3 2030	2,173.944	Model
Q4 2030	2,564.675	Model

Quarterly revenue-vector derivation

All rows below are reference inputs. Gross profit = vector revenue x the stated gross margin. The next table applies the separately weighted evidence deltas. A product-level proxy is a model assumption even when the group total is calibrated to guidance.

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2026	Optical systems and external components	980.0	42.0%	411.6
Q3 2026	IP routing and data-center switching	770.0	43.0%	331.1
Q3 2026	Fixed access networks	510.0	44.0%	224.4
Q3 2026	Radio networks	1,850.0	31.0%	573.5
Q3 2026	Core software	445.0	60.0%	267.0
Q3 2026	Technology-standard licensing	390.0	98.0%	382.2
Q3 2026	Remaining portfolio and eliminations	95.0	28.0%	26.6
Q4 2026	Optical systems and external components	1,120.0	44.0%	492.8
Q4 2026	IP routing and data-center switching	920.0	46.0%	423.2
Q4 2026	Fixed access networks	615.0	45.0%	276.8
Q4 2026	Radio networks	2,300.0	40.0%	920.0
Q4 2026	Core software	670.0	70.0%	469.0
Q4 2026	Technology-standard licensing	310.0	98.0%	303.8
Q4 2026	Remaining portfolio and eliminations	105.0	30.0%	31.5
Q1 2027	Optical systems and external components	1,012.0	45.0%	455.4
Q1 2027	IP routing and data-center switching	735.0	45.5%	334.4
Q1 2027	Fixed access networks	462.0	45.5%	210.2
Q1 2027	Radio networks	1,650.2	33.0%	544.6
Q1 2027	Core software	552.2	65.0%	359.0
Q1 2027	Technology-standard licensing	387.5	98.0%	379.8
Q1 2027	Remaining portfolio and eliminations	97.5	30.0%	29.2
Q2 2027	Optical systems and external components	1,104.0	45.0%	496.8
Q2 2027	IP routing and data-center switching	805.0	45.5%	366.3
Q2 2027	Fixed access networks	528.0	45.5%	240.2
Q2 2027	Radio networks	1,891.8	34.0%	643.2
Q2 2027	Core software	575.8	66.0%	380.0
Q2 2027	Technology-standard licensing	387.5	98.0%	379.8
Q2 2027	Remaining portfolio and eliminations	97.5	30.0%	29.2

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2027	Optical systems and external components	1,150.0	45.0%	517.5
Q3 2027	IP routing and data-center switching	875.0	45.5%	398.1
Q3 2027	Fixed access networks	550.0	45.5%	250.2
Q3 2027	Radio networks	1,972.2	32.0%	631.1
Q3 2027	Core software	493.5	62.0%	306.0
Q3 2027	Technology-standard licensing	387.5	98.0%	379.8
Q3 2027	Remaining portfolio and eliminations	97.5	30.0%	29.2
Q4 2027	Optical systems and external components	1,334.0	45.0%	600.3
Q4 2027	IP routing and data-center switching	1,085.0	45.5%	493.7
Q4 2027	Fixed access networks	660.0	45.5%	300.3
Q4 2027	Radio networks	2,535.8	41.0%	1,039.7
Q4 2027	Core software	728.5	72.0%	524.5
Q4 2027	Technology-standard licensing	387.5	98.0%	379.8
Q4 2027	Remaining portfolio and eliminations	97.5	30.0%	29.2
Q1 2028	Optical systems and external components	1,276.0	46.5%	593.3
Q1 2028	IP routing and data-center switching	861.0	46.5%	400.4
Q1 2028	Fixed access networks	483.0	46.0%	222.2
Q1 2028	Radio networks	1,670.8	34.0%	568.1
Q1 2028	Core software	580.4	66.0%	383.1
Q1 2028	Technology-standard licensing	400.0	98.0%	392.0
Q1 2028	Remaining portfolio and eliminations	90.0	31.0%	27.9
Q2 2028	Optical systems and external components	1,392.0	46.5%	647.3
Q2 2028	IP routing and data-center switching	943.0	46.5%	438.5
Q2 2028	Fixed access networks	552.0	46.0%	253.9
Q2 2028	Radio networks	1,915.2	35.0%	670.3
Q2 2028	Core software	605.1	67.0%	405.5
Q2 2028	Technology-standard licensing	400.0	98.0%	392.0
Q2 2028	Remaining portfolio and eliminations	90.0	31.0%	27.9
Q3 2028	Optical systems and external components	1,450.0	46.5%	674.2
Q3 2028	IP routing and data-center switching	1,025.0	46.5%	476.6
Q3 2028	Fixed access networks	575.0	46.0%	264.5

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2028	Radio networks	1,996.8	33.0%	658.9
Q3 2028	Core software	518.7	63.0%	326.8
Q3 2028	Technology-standard licensing	400.0	98.0%	392.0
Q3 2028	Remaining portfolio and eliminations	90.0	31.0%	27.9
Q4 2028	Optical systems and external components	1,682.0	46.5%	782.1
Q4 2028	IP routing and data-center switching	1,271.0	46.5%	591.0
Q4 2028	Fixed access networks	690.0	46.0%	317.4
Q4 2028	Radio networks	2,567.2	42.0%	1,078.2
Q4 2028	Core software	765.7	73.0%	559.0
Q4 2028	Technology-standard licensing	400.0	98.0%	392.0
Q4 2028	Remaining portfolio and eliminations	90.0	31.0%	27.9
Q1 2029	Optical systems and external components	1,518.0	47.5%	721.0
Q1 2029	IP routing and data-center switching	955.5	47.0%	449.1
Q1 2029	Fixed access networks	497.7	46.0%	228.9
Q1 2029	Radio networks	1,681.0	34.5%	579.9
Q1 2029	Core software	603.9	66.5%	401.6
Q1 2029	Technology-standard licensing	412.5	98.0%	404.2
Q1 2029	Remaining portfolio and eliminations	85.0	31.0%	26.4
Q2 2029	Optical systems and external components	1,656.0	47.5%	786.6
Q2 2029	IP routing and data-center switching	1,046.5	47.0%	491.9
Q2 2029	Fixed access networks	568.8	46.0%	261.6
Q2 2029	Radio networks	1,927.0	35.5%	684.1
Q2 2029	Core software	629.6	67.5%	425.0
Q2 2029	Technology-standard licensing	412.5	98.0%	404.2
Q2 2029	Remaining portfolio and eliminations	85.0	31.0%	26.4
Q3 2029	Optical systems and external components	1,725.0	47.5%	819.4
Q3 2029	IP routing and data-center switching	1,137.5	47.0%	534.6
Q3 2029	Fixed access networks	592.5	46.0%	272.6
Q3 2029	Radio networks	2,009.0	33.5%	673.0
Q3 2029	Core software	539.7	63.5%	342.7
Q3 2029	Technology-standard licensing	412.5	98.0%	404.2

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2029	Remaining portfolio and eliminations	85.0	31.0%	26.4
Q4 2029	Optical systems and external components	2,001.0	47.5%	950.5
Q4 2029	IP routing and data-center switching	1,410.5	47.0%	662.9
Q4 2029	Fixed access networks	711.0	46.0%	327.1
Q4 2029	Radio networks	2,583.0	42.5%	1,097.8
Q4 2029	Core software	796.7	73.5%	585.6
Q4 2029	Technology-standard licensing	412.5	98.0%	404.2
Q4 2029	Remaining portfolio and eliminations	85.0	31.0%	26.4
Q1 2030	Optical systems and external components	1,694.0	48.0%	813.1
Q1 2030	IP routing and data-center switching	1,029.0	47.0%	483.6
Q1 2030	Fixed access networks	504.0	46.0%	231.8
Q1 2030	Radio networks	1,691.2	34.5%	583.5
Q1 2030	Core software	627.4	66.5%	417.3
Q1 2030	Technology-standard licensing	425.0	98.0%	416.5
Q1 2030	Remaining portfolio and eliminations	80.0	31.0%	24.8
Q2 2030	Optical systems and external components	1,848.0	48.0%	887.0
Q2 2030	IP routing and data-center switching	1,127.0	47.0%	529.7
Q2 2030	Fixed access networks	576.0	46.0%	265.0
Q2 2030	Radio networks	1,938.8	35.5%	688.3
Q2 2030	Core software	654.1	67.5%	441.6
Q2 2030	Technology-standard licensing	425.0	98.0%	416.5
Q2 2030	Remaining portfolio and eliminations	80.0	31.0%	24.8
Q3 2030	Optical systems and external components	1,925.0	48.0%	924.0
Q3 2030	IP routing and data-center switching	1,225.0	47.0%	575.8
Q3 2030	Fixed access networks	600.0	46.0%	276.0
Q3 2030	Radio networks	2,021.2	33.5%	677.1
Q3 2030	Core software	560.7	63.5%	356.0
Q3 2030	Technology-standard licensing	425.0	98.0%	416.5
Q3 2030	Remaining portfolio and eliminations	80.0	31.0%	24.8
Q4 2030	Optical systems and external components	2,233.0	48.0%	1,071.8
Q4 2030	IP routing and data-center switching	1,519.0	47.0%	713.9

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q4 2030	Fixed access networks	720.0	46.0%	331.2
Q4 2030	Radio networks	2,598.8	42.5%	1,104.5
Q4 2030	Core software	827.7	73.5%	608.4
Q4 2030	Technology-standard licensing	425.0	98.0%	416.5
Q4 2030	Remaining portfolio and eliminations	80.0	31.0%	24.8

Operating-cost derivation

Quarter	R&D including SBC	SG&A and net other operating costs including SBC	Restructuring charges	Acquired-intangible amortization	Integration and impairment reserve
Q3 2026	1,150.0	631.4	154.0	46.0	15.0
Q4 2026	1,175.0	512.1	100.0	46.0	15.0
Q1 2027	1,101.7	548.4	60.0	43.8	11.2
Q2 2027	1,212.6	603.6	60.0	43.8	11.2
Q3 2027	1,243.3	618.9	60.0	43.8	11.2
Q4 2027	1,536.4	764.8	60.0	43.8	11.2
Q1 2028	1,158.0	579.0	40.0	41.2	7.5
Q2 2028	1,273.8	636.9	40.0	41.2	7.5
Q3 2028	1,308.0	654.0	40.0	41.2	7.5
Q4 2028	1,612.6	806.3	40.0	41.2	7.5
Q1 2029	1,191.0	598.4	30.0	37.5	5.0
Q2 2029	1,309.4	657.8	30.0	37.5	5.0
Q3 2029	1,345.7	676.1	30.0	37.5	5.0
Q4 2029	1,655.9	832.0	30.0	37.5	5.0
Q1 2030	1,222.2	617.2	30.0	33.8	5.0
Q2 2030	1,343.1	678.2	30.0	33.8	5.0
Q3 2030	1,381.1	697.4	30.0	33.8	5.0
Q4 2030	1,697.5	857.2	30.0	33.8	5.0

The categories reconcile to reference operating expenses. Incremental costs below are applied once; depreciation or stock compensation already included in a reference category is not added again.

Probability-weighted financial bridge

Active = reference + sum(probability x bounded conditional delta). Negative deltas are retained. A branch with zero near-term increment remains zero only for the timing or overlap reasons documented in the evidence register.

Weighted OP below is after the stated minority operating-profit attribution. Conditional gross profit and expenses remain consolidated figures.

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2026	OPTICAL QUALIFIED OUTPUT	55%	0.0	0.0	0.0	0.0
Q3 2026	REPRICEABLE OPTICAL	30%	0.0	0.0	0.0	0.0
Q3 2026	AI RAN CONVERSION	35%	0.0	0.0	0.0	0.0
Q3 2026	RESIDUAL INPUT COST	40%	0.0	0.0	0.0	0.0
Q4 2026	OPTICAL QUALIFIED OUTPUT	55%	0.0	0.0	0.0	0.0
Q4 2026	REPRICEABLE OPTICAL	30%	0.0	0.0	0.0	0.0
Q4 2026	AI RAN CONVERSION	35%	0.0	0.0	0.0	0.0
Q4 2026	RESIDUAL INPUT COST	40%	0.0	0.0	0.0	0.0
Q1 2027	OPTICAL QUALIFIED OUTPUT	55%	40.0	16.0	4.8	6.1
Q1 2027	REPRICEABLE OPTICAL	30%	5.1	3.0	0.3	0.8
Q1 2027	AI RAN CONVERSION	35%	10.0	3.5	2.0	0.5
Q1 2027	RESIDUAL INPUT COST	40%	0.0	0.0	17.7	-7.0
Q2 2027	OPTICAL QUALIFIED OUTPUT	55%	80.0	32.0	9.6	12.2
Q2 2027	REPRICEABLE OPTICAL	30%	5.5	3.3	0.3	0.9
Q2 2027	AI RAN CONVERSION	35%	20.0	7.0	4.0	1.0
Q2 2027	RESIDUAL INPUT COST	40%	0.0	0.0	19.5	-7.7
Q3 2027	OPTICAL QUALIFIED OUTPUT	55%	120.0	48.0	14.4	18.3
Q3 2027	REPRICEABLE OPTICAL	30%	5.8	3.4	0.3	0.9
Q3 2027	AI RAN CONVERSION	35%	30.0	10.5	6.0	1.6
Q3 2027	RESIDUAL INPUT COST	40%	0.0	0.0	20.6	-8.2
Q4 2027	OPTICAL QUALIFIED OUTPUT	55%	160.0	64.0	19.2	24.4
Q4 2027	REPRICEABLE OPTICAL	30%	6.7	4.0	0.3	1.1
Q4 2027	AI RAN CONVERSION	35%	40.0	14.0	8.0	2.1
Q4 2027	RESIDUAL INPUT COST	40%	0.0	0.0	24.6	-9.8

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q1 2028	OPTICAL QUALIFIED OUTPUT	55%	80.0	32.0	9.6	12.2
Q1 2028	REPRICEABLE OPTICAL	30%	6.4	3.8	0.3	1.0
Q1 2028	AI RAN CONVERSION	35%	30.0	10.5	6.0	1.6
Q1 2028	RESIDUAL INPUT COST	40%	0.0	0.0	21.0	-8.3
Q2 2028	OPTICAL QUALIFIED OUTPUT	55%	160.0	64.0	19.2	24.4
Q2 2028	REPRICEABLE OPTICAL	30%	7.0	4.2	0.3	1.1
Q2 2028	AI RAN CONVERSION	35%	60.0	21.0	12.0	3.1
Q2 2028	RESIDUAL INPUT COST	40%	0.0	0.0	23.1	-9.1
Q3 2028	OPTICAL QUALIFIED OUTPUT	55%	240.0	96.0	28.8	36.6
Q3 2028	REPRICEABLE OPTICAL	30%	7.2	4.3	0.4	1.2
Q3 2028	AI RAN CONVERSION	35%	90.0	31.5	18.0	4.7
Q3 2028	RESIDUAL INPUT COST	40%	0.0	0.0	24.4	-9.7
Q4 2028	OPTICAL QUALIFIED OUTPUT	55%	320.0	128.0	38.4	48.8
Q4 2028	REPRICEABLE OPTICAL	30%	8.4	5.0	0.4	1.4
Q4 2028	AI RAN CONVERSION	35%	120.0	42.0	24.0	6.2
Q4 2028	RESIDUAL INPUT COST	40%	0.0	0.0	29.1	-11.5
Q1 2029	OPTICAL QUALIFIED OUTPUT	55%	120.0	48.0	14.4	18.3
Q1 2029	REPRICEABLE OPTICAL	30%	5.7	3.4	0.3	0.9
Q1 2029	AI RAN CONVERSION	35%	60.0	21.0	12.0	3.1
Q1 2029	RESIDUAL INPUT COST	40%	0.0	0.0	23.8	-9.4
Q2 2029	OPTICAL QUALIFIED OUTPUT	55%	240.0	96.0	28.8	36.6
Q2 2029	REPRICEABLE OPTICAL	30%	6.2	3.7	0.3	1.0
Q2 2029	AI RAN CONVERSION	35%	120.0	42.0	24.0	6.2
Q2 2029	RESIDUAL INPUT COST	40%	0.0	0.0	26.2	-10.4
Q3 2029	OPTICAL QUALIFIED OUTPUT	55%	360.0	144.0	43.2	54.9
Q3 2029	REPRICEABLE OPTICAL	30%	6.5	3.9	0.3	1.1
Q3 2029	AI RAN CONVERSION	35%	180.0	63.0	36.0	9.4

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2029	RESIDUAL INPUT COST	40%	0.0	0.0	27.6	-10.9
Q4 2029	OPTICAL QUALIFIED OUTPUT	55%	480.0	192.0	57.6	73.2
Q4 2029	REPRICEABLE OPTICAL	30%	7.5	4.5	0.4	1.2
Q4 2029	AI RAN CONVERSION	35%	240.0	84.0	48.0	12.5
Q4 2029	RESIDUAL INPUT COST	40%	0.0	0.0	33.0	-13.1
Q1 2030	OPTICAL QUALIFIED OUTPUT	55%	150.0	60.0	18.0	22.9
Q1 2030	REPRICEABLE OPTICAL	30%	4.2	2.5	0.2	0.7
Q1 2030	AI RAN CONVERSION	35%	90.0	31.5	18.0	4.7
Q1 2030	RESIDUAL INPUT COST	40%	0.0	0.0	25.8	-10.2
Q2 2030	OPTICAL QUALIFIED OUTPUT	55%	300.0	120.0	36.0	45.7
Q2 2030	REPRICEABLE OPTICAL	30%	4.6	2.8	0.2	0.8
Q2 2030	AI RAN CONVERSION	35%	180.0	63.0	36.0	9.4
Q2 2030	RESIDUAL INPUT COST	40%	0.0	0.0	28.4	-11.2
Q3 2030	OPTICAL QUALIFIED OUTPUT	55%	450.0	180.0	54.0	68.6
Q3 2030	REPRICEABLE OPTICAL	30%	4.8	2.9	0.2	0.8
Q3 2030	AI RAN CONVERSION	35%	270.0	94.5	54.0	14.0
Q3 2030	RESIDUAL INPUT COST	40%	0.0	0.0	30.0	-11.9
Q4 2030	OPTICAL QUALIFIED OUTPUT	55%	600.0	240.0	72.0	91.5
Q4 2030	REPRICEABLE OPTICAL	30%	5.6	3.3	0.3	0.9
Q4 2030	AI RAN CONVERSION	35%	360.0	126.0	72.0	18.7
Q4 2030	RESIDUAL INPUT COST	40%	0.0	0.0	35.8	-14.2

Consolidated profit to the common shareholder claim

Minority attribution is an earnings allocation, not operating expense or a fixed book-value deduction. Its rate is a disclosed model proxy and moves with reference profit and each probability-weighted change.

Quarter	Consolidated OP	Minority rate	Minority allocation	Owner OP
Q3 2026	220.0	1.0%	2.2	217.8
Q4 2026	1,069.0	1.0%	10.7	1,058.3
Q1 2027	547.9	1.0%	5.5	542.4
Q2 2027	610.7	1.0%	6.1	604.6
Q3 2027	547.6	1.0%	5.5	542.1
Q4 2027	969.3	1.0%	9.7	959.6
Q1 2028	767.7	1.0%	7.7	760.0
Q2 2028	855.6	1.0%	8.6	847.0
Q3 2028	803.4	1.0%	8.0	795.4
Q4 2028	1,285.2	1.0%	12.9	1,272.4
Q1 2029	962.4	1.0%	9.6	952.8
Q2 2029	1,073.9	1.0%	10.7	1,063.2
Q3 2029	1,033.4	1.0%	10.3	1,023.1
Q4 2029	1,568.6	1.0%	15.7	1,552.9
Q1 2030	1,080.7	1.0%	10.8	1,069.9
Q2 2030	1,207.8	1.0%	12.1	1,195.8
Q3 2030	1,175.3	1.0%	11.8	1,163.5
Q4 2030	1,745.6	1.0%	17.5	1,728.2

Enterprise-to-equity reconciliation

20x rolling Q+next3 IFRS operating profit after1% common-attribution reserve, multiplied by1.14 USD/EUR and divided by window-end ADS-equivalent diluted shares; no excess-cash bonus.

Quarter	Reference claims	Weighted change	Total claims	Ceiling / share
Q3 2026	0.0	0.0	0.0	9.70
Q4 2026	0.0	0.0	0.0	10.99
Q1 2027	0.0	0.0	0.0	10.58
Q2 2027	0.0	0.0	0.0	11.43
Q3 2027	0.0	0.0	0.0	12.38
Q4 2027	0.0	0.0	0.0	13.37
Q1 2028	0.0	0.0	0.0	14.59
Q2 2028	0.0	0.0	0.0	15.34
Q3 2028	0.0	0.0	0.0	16.17
Q4 2028	0.0	0.0	0.0	17.05
Q1 2029	0.0	0.0	0.0	18.13
Q2 2029	0.0	0.0	0.0	18.57
Q3 2029	0.0	0.0	0.0	19.07
Q4 2029	0.0	0.0	0.0	19.59
Q1 2030	0.0	0.0	0.0	20.25
Q2 2030	0.0	0.0	0.0	-
Q3 2030	0.0	0.0	0.0	-
Q4 2030	0.0	0.0	0.0	-

Claim decisions, overlap and triggers

OPTICAL_QUALIFIED_OUTPUT

source: Q1/Q2 management and Q2 capacity note; OFC roadmap

date: 2026-07-23

p: 0.55

hypothesis: Faster qualified optical conversion above the already growing reference

reference: Optical external sales EUR4.6/5.8/6.9/7.7bn in 2027-30 already include San Jose and normal replacement growth.

Conditional: Additional EUR0.4/0.8/1.2/1.5bn external annual invoices, phased 10/20/30/40%; proxies for accepted good output, not wafer capacity guidance.

derivation: At EUR20k external-system-equivalent ASP: incremental 20k/40k/60k/75k equivalents; gross margin40% after production D&A/rejects/expediting, incremental R&D and selling12%.

overlap: The EUR2.8bn AI order intake is not added again; it includes IP and existing backlog. Internal components are eliminated. Volume branch excludes price branch.

counter: Reduce to30% if San Jose output/qualification delays by two quarters or cancellations; raise toward70% only on accepted sales and yields.

REPRICEABLE_OPTICAL

source: April management pricing caution plus July elongated lead times

date: 2026-07-23

p: 0.3

hypothesis: Limited premium on new/repriceable optical orders

reference: Volume-driven growth and normal cost decline already present. No premium on fixed orders.

Conditional: 2% premium on25% of reference optical sales in2027-28;1.5% in2029;1% in2030. Plausible range0-3%.

derivation: 60% of price increase retained in GP after materials/overtime;5% additional commercial expense; no unit uplift.

overlap: Only reference units; incremental qualified-output branch receives no added price premium.

counter: April CEO explicitly said growth was volume-led, not price-led. Reduce to10% if terms remain fixed or cost inflation absorbs increase.

AI_RAN_CONVERSION

source: Q2 management pilot and commercial schedule

date: 2026-07-23

p: 0.35

hypothesis: Net additional AI-RAN software/platform sales after displacement of conventional RAN

reference: Radio sales near-flat, with ongoing software upgrade and R&D already modeled.

Conditional: Net additional EUR100/300/600/900m in2027-30, equivalent to10/20/30/30 programs at EUR10/15/20/30m annual net Nokia revenue.

derivation: 35% GP after hardware and deployment costs,20% incremental R&D/support; USD GPU vendor revenue excluded.

overlap: Net of replacement radio sales; pilot customer count does not mean commercial contracts.

counter: Reduce to15% on pilot slips or no measured operator economics; raise to50% on paid production programs.

RESIDUAL_INPUT_COST

source: July management memory and supply-chain commentary

date: 2026-07-23

p: 0.4

hypothesis: Unrecovered input costs above normal production and branch expenses

reference: Ordinary material cost, yield loss and depreciation included in each vector GM.

Conditional: Additional0.8% of reference Network Infrastructure revenue as operating cost from2027; sensitivity0-1.5%.

derivation: Remaining memory/DSP/PCB procurement expense after normal mitigation and pass-through; affects margin only.

overlap: Does not duplicate optical branch expediting or recurring semiconductor bill.

counter: Remove as realized recovered pricing demonstrably offsets the shock; increase if shortages persist without pass-through.

Remaining bounded uncertainties

- Per-product gross margins are calibrated assumptions, not issuer disclosures; only segment GMs are disclosed.

- Chandler acquisition price, conversion capex and precise production acceptance date are undisclosed; cash sensitivities reserve explicit amounts.

- Product units/ASP are equivalent-dollar proxies; actual good wafer output, internal/external mix and customer terms are not disclosed.

- EUR/USD1.14 is a modeling convention drawn from July guidance, not a promised future or current spot rate.

- One-percent OP minority proxy bounds small ongoing NCI, but exact subsidiary profit and cash attribution are not public.

- Q1 full transcript was inspected via web; direct archive failed403. Latest completed quarter isQ2; October22 results are pending.

Sources and basis

The current production framework was read before initiation. Source statements are evidence, not instructions. The shared model data drives both this PDF and the linked Excel model; the clean-state editable artifact embeds these same images.

[1] Primary source | 2026-07-23 | https://www.nokia.com/system/files/2026-07/nokia_results_2026_q2.pdf | Archived SHA-256: 6792b45b61c9512332f296bcd02e97704768a24e5153082586f282ad9771d505

[2] Primary source | 2026-07-23 | https://www.nokia.com/system/files/2026-07/nokia_slides_2026_q2.pdf | Archived SHA-256: 2f87eb2cd476060fb603c590bac107b4391797f33178ed1fae4ad219493ba6a6

[3] Primary source | 2026-04-23 | https://www.nokia.com/system/files/2026-04/nokia_results_2026_q1.pdf | Archived SHA-256: 6b2f1031b46485a899476d780b2ff3c42eb48c5bd6a0716c314e0f5a5a3d8df3

[4] Primary source | 2026-04-23 | https://www.nokia.com/system/files/2026-04/nokia_slides_2026_q1.pdf | Archived SHA-256: 62225e8ec2eb7aa00df73ff27b3fc951e0647a7f8f491b887c22b64edc0e4f43

[5] Primary source | 2026-03-05 | https://www.nokia.com/system/files/2026-03/nokia-form-20-f-2025.pdf | Archived SHA-256: e26a2034501e2bfb866d03cfee93e535cec926e094c610bc3dff4dc4c600fd57

[6] Primary source | 2026-07-23 | https://www.fool.com/earnings/call-transcripts/2026/07/23/nokia-nok-q2-2026-earnings-call-transcript/ | Archived SHA-256: 985485a3e0ebf5154a3a6abb9504de79ce338a57808847643d219aa977c1f7db

[7] Primary source | 2026-04-23 | https://www.roic.ai/quote/NOK%3ATB/transcripts/2026-year/1-quarter

[8] Primary source | 2026-10-07 accessed | https://www.nokia.com/about-us/investors/stock-information/number-of-shares/ | Archived SHA-256: 8e25a6a303b18d97a96bea69a5bc994d3d3bd997807b16742516823a2197bf83

[9] Primary source | 2026-03-18 | https://www.nokia.com/sites/default/files/2026-03/2026-ofc-event-18-march-2026.pdf | Archived SHA-256: 2f814de26463dcf5432822c28e484e748c23e3807b75ad66a1c0bcc25302599c

[10] Primary source | 2026-04-23 | https://www.gurufocus.com/stock/NOK/transcripts/8814473

[11] Primary source | 2026-10-07 accessed | https://www.nokia.com/optical-networks/ice-d-intra-datacenter/ | Archived SHA-256: d7d5569da25130ddd87beca27ff086b835d431ea90acc9d8a153e0afaa6a000b

[12] Primary source | 2026-10-07 accessed | https://www.nokia.com/optical-networks/ice-d-1600g-dr8-lpo/ | Archived SHA-256: bc15de4afc574ba64f4df2a4944c8a0f21a763d9651266ca15e8b38a40064569

[13] Primary source | 2026-04-23 | https://www.roic.ai/quote/NOK/transcripts/2026-year/1-quarter